

K-Bound: Selective Test-Time Adaptation

ImageNet-C (noise) results brief — when a gate on TTA actually helps

ResNet-50 · device MPS · alpha = 0.1 · 36 conditions/method (3 corruptions x 3 severities x 4 batch×aggressiveness regimes) · TTA methods: TENT, EATA, SAR (with collapse-recovery fix) · generated 2026-06-09

Summary

K-Bound is a per-condition gate that decides whether to apply test-time adaptation (TTA), freeze, or abstain. The headline finding is clean and honest: **the gate is essentially free when adaptation is safe, and it strictly wins when adaptation is risky**. It matches or ties the best naive policy for TENT and EATA (where adapting almost always helps), and on SAR — where adapting hurts 31% of the time and is net-negative on average — it beats *both* ‘always-adapt’ and ‘always-freeze’, cutting regret to oracle by ~3x versus the better baseline. The advantage switches on exactly when the harmful fraction crosses ~0.1.

Method	Harmful %	Regret: always-adapt	Regret: always-freeze	Regret: K-Bound	Oracle acc	Beats both?
TENT	8%	0.0003	0.0440	0.0042	0.454	no
EATA	3%	0.0001	0.0349	0.0003	0.444	no
SAR	31%	0.0606	0.0277	0.0086	0.437	YES

Table 1. Regret to oracle (lower is better). K-Bound is the only policy that is never worst and that beats both baselines once adaptation turns risky (SAR).

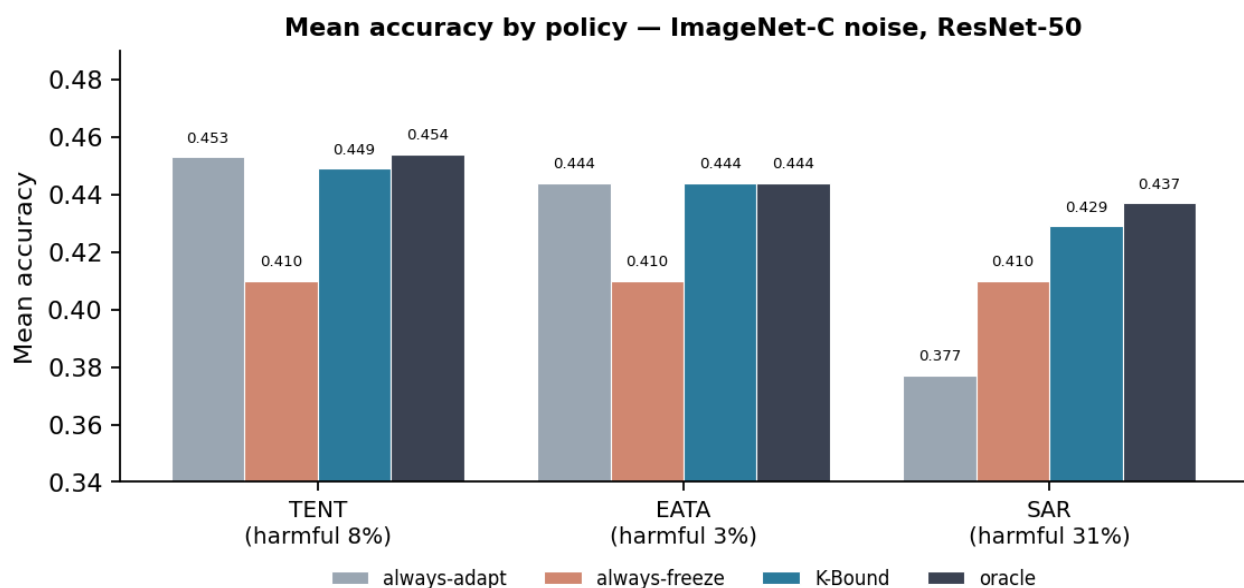


Figure 1. K-Bound (teal) tracks the oracle across all three methods. On SAR, ‘always-adapt’ collapses below even ‘always-freeze’; K-Bound stays near oracle.

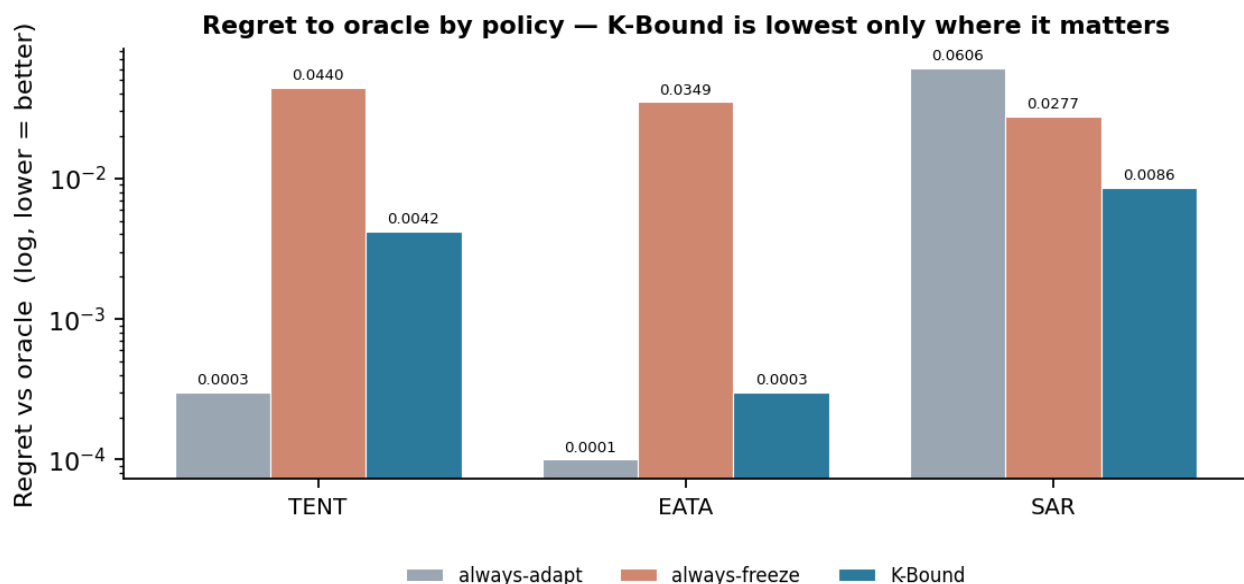


Figure 2. Regret to oracle (log scale). Where adaptation is safe (TENT, EATA) the gate costs a hair; where it is dangerous (SAR) the gate cuts regret far below both baselines.

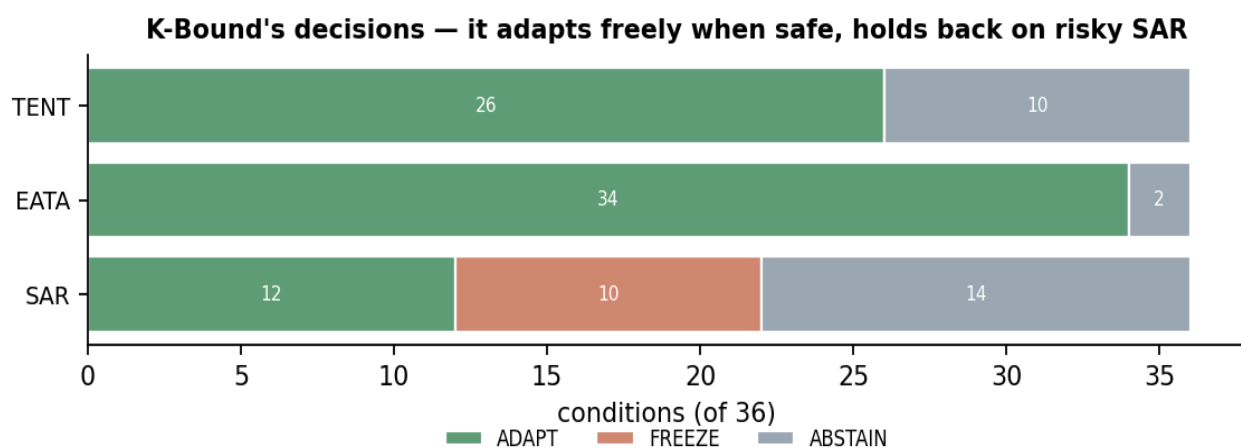


Figure 3. The gate's behavior explains the result: it adapts on 34/36 EATA conditions, but freezes or abstains on 24/36 SAR conditions.

What this shows

The value of a selective TTA gate depends entirely on how often the underlying method is harmful. For TENT (8% harmful) and EATA (3%), adapting is nearly always safe, so 'always-adapt' is already near-oracle and there is little to gain — K-Bound ties it and, critically, does no harm (regret 0.0003–0.0042). For SAR (31% harmful, mean benefit negative), neither naive policy is good, and K-Bound's selectivity is what recovers most of the oracle accuracy (0.429 vs oracle 0.437; vs 0.377 always-adapt and 0.410 always-freeze). Across all three methods the crossover is consistent: **K-Bound strictly beats both baselines once the harmful fraction is $\geq \sim 0.1$.**

Limitations & what would make this a flagship claim

This is a real, honest result but a narrow one: a single dataset family (ImageNet-C noise), one backbone (ResNet-50), 36 conditions/method, and the decisive win rests on one method (SAR). A reviewer's fair objection is 'you are mainly rescuing an unstable method.' To turn the ~ 0.1 crossover from an observation into a *general law*, the next runs should populate the harmful-fraction axis with many points — all 15 ImageNet-C corruptions, a ViT backbone alongside ResNet-50, plus the ImageNet-R and Camelyon17 sweeps — and plot regret-reduction versus

harmful fraction on one figure. If the turn-on holds across datasets, architectures, and methods, that single plot is the paper's contribution.

Source: `experiments/kbound/results/imagenetc_noise/decisive_tta_results.json` (`cifar_tent_mps_v2.py`, `kbtrain.sh noise-fast`). All numbers measured, not estimated. SAR uses the collapse-recovery fix (restore weights+optimizer, reset EMA).